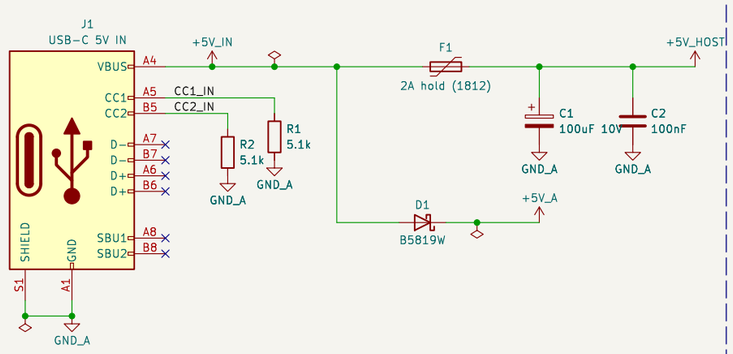
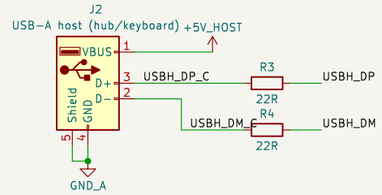
Power input: 5 V adapter on USB-C (power only)

Adapter 5 V, 2–3 A. F1 protects the USB-A port; D1 feeds MCU-A so the keyboard works while PC 1 is off.



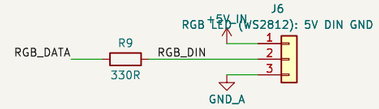
USB host port: keyboard, mouse, dongle or hub

USBH_DP / USBH_DM come from MCU-A GP2 / GP3 (PIO-USB host).



Optional RGB LED header J6 (WS2812 / SK6812)

Data from GP6 through R9. Most WS2812B accept 3.3 V data at 5 V; keep the wire short.

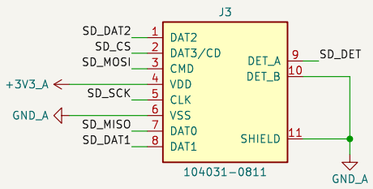


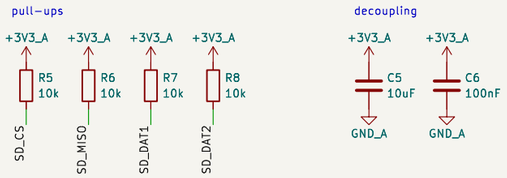
Mounting holes (M2.5)



microSD: shared drive (SPI1)

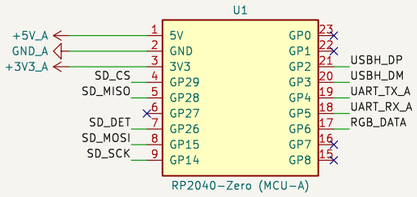
Shown as a USB drive on the active PC only.
10k pull-ups on CS, MISO and the unused DAT1 / DAT2.





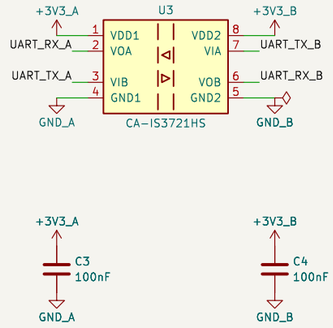
MCU-A: RP2040-Zero (to PC 1 by its own USB-C)

USB host, router and microSD.
On-board WS2812 on GP16 shows the active PC.

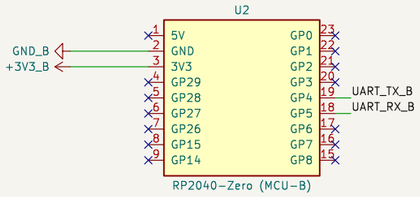


Isolation: GND_A and GND_B never meet

CA-IS3721HS, pin compatible with ADuM1201.
Use a >= 10 Mbps grade for a 3 Mbaud UART.



MCU-B: RP2040-Zero (to PC 2, powered by PC 2)



Notes

GPIO map (same firmware for both variants)
- GP2 / GP3 — USB host D+ / D− (PIO-USB, 22 ohm series)
- GP4 / GP5 — UART1 TX / RX through the isolator
- GP14 / GP15 — SPI1 SCK / MOSI to microSD
- GP28 / GP29 — SPI1 MISO / card CS
- GP26 — card detect (switch to GND, internal pull-up)
- GP6 — external RGB LED data (header J6)
- GP16 — on-board WS2812 of the RP2040-Zero (status)

Power domains
- side A (+5V_IN, +5V_HOST, +5V_A, +3V3_A, GND_A): adapter on J1, MCU-A, USB host port, microSD. PC 1 connects to U1's USB-C.
- side B (+3V3_B, GND_B): MCU-B, powered by PC 2 through U2's USB-C. The only link between the sides is U3.

kmswitch (open hardware)		
Sheet: /		
File: kmswitch-a.kicad_sch		
Title: kmswitch — variant A (RP2040-Zero carrier)		
Size: A3	Date:	**Rev: 1.0**
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